Standard Well Log Display Examples

Display types, curves, positioning, scaling recommendations, and usage

Purpose. This reference summarizes common well-log display templates for a Well Log Viewer. The emphasis is on practical display structure: which curves belong together, where tracks are normally positioned, how curves are commonly scaled, and when each display is used. Exact curve names and scales vary by service company, vintage, units, basin, tool generation, and local interpretation practice.

Design principle. The viewer should not be built around line curves only. A durable WLV needs curve tracks, image tracks, waveform tracks, categorical tracks, event tracks, annotation tracks, schematic tracks, and computed-interpretation tracks.

1. Recommended WLV Template Library

Template	Core display content	Primary usage
Basic Lithology	Depth, GR/SP, caliper/bit size, formation tops	Quick lithology review, correlation, shale/sand screening
Open-Hole Triple Combo	GR/SP/caliper + resistivity + density/neutron	Default petroleum formation evaluation
Open-Hole Quad Combo	Triple combo plus sonic	Expanded formation evaluation and seismic tie preparation
Reservoir Evaluation	Triple combo + Vsh + PHIE/PHIT + Sw + net pay flags	Petrophysical interpretation and pay review
Resistivity / Invasion Review	GR plus shallow/medium/deep resistivity grouped by investigation depth	Fluid indication, invasion diagnosis, saturation review
Density-Neutron Lithology	GR + RHOB/NPHI + PEF + DRHO	Porosity, gas crossover, lithology discrimination
Sonic / Seismic Tie	GR + sonic + density + impedance + synthetic/reflectivity	Well-to-seismic calibration and rock physics
Image Log Review	GR + borehole image + dip/fracture/facies tracks	Fracture, bedding, facies, borehole condition
Geochemical / Mineralogy	Spectral GR, elemental yields, mineral volumes	Mineralogy, unconventional reservoir evaluation
NMR Evaluation	NMR porosity, free/bound fluid, T2 distribution, permeability	Movable fluid, pore-size and permeability review
Cased-Hole Cement Bond	GR/CCL + CBL amplitude + VDL + bond index/cement map	Cement and zonal-isolation evaluation
Cased-Hole Saturation	GR/CCL + sigma/C-O/ratios + interpreted saturation	Behind-casing saturation monitoring
Production Logging	GR/CCL + spinner + pressure + temperature + holdup + completion schematic	Flow allocation, leaks, crossflow, entry-point diagnosis
Well Integrity / Casing Inspection	GR/CCL + caliper/radius/thickness + casing image	Corrosion, deformation, perforation damage, casing review
Core Integration	Logs + core photos + plug porosity/permeability + facies	Log-core calibration and facies validation
Completion / Intervention Review	GR/CCL + completion schematic + perforations + pressure/temp/flow	Workover, intervention, and completion review

2. Display Types, Track Positioning, Curves, Scaling, and Usage

Open-Hole Triple Combo Display

The baseline petroleum well-log display. This should be the first Auto-Build template in the WLV.

Track position	Curves / data	Recommended scaling	Typical usage
Track 0 / reference	Depth, MD/TVD, formation tops, casing shoe, core points	Linear depth scale. Common vertical scales include 1:200, 1:500, 1:1000, 5 in/100 ft, and 2 in/100 ft.	Main alignment axis and interpretation reference.

Track 1 / left	GR, SP, caliper, bit size	GR 0-150 API or 0-200 API. SP linear mV, often reversed or centered. Caliper linear in inches or mm.	Lithology, shale/sand discrimination, borehole quality.
Track 2 / middle	Shallow, medium, and deep resistivity	Logarithmic. Common ranges: 0.2-20, 0.2-200, or 0.2-2000 ohm-m.	Fluid indication, invasion profile, hydrocarbon/water contrast.
Track 3 / right	RHOB, NPHI, sometimes PEF	RHOB commonly 1.95-2.95 g/cc or 1.65-2.95 g/cc. NPHI commonly 45 to -15 p.u. reversed, or 0.45 to -0.15.	Porosity, lithology, gas effect, density-neutron crossover.

Open-Hole Quad Combo / Full Evaluation Display

Triple combo plus sonic. Use when sonic curves are available.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, SP, caliper, bit size	Same as triple combo.	Lithology and borehole condition.
Track 2	MSFL, LLS, LLD, ILD, AT10/AT30/AT90, RT, RXO	Log scale, usually 0.2-200 or 0.2-2000 ohm-m.	Saturation, invasion, reservoir discrimination.
Track 3	RHOB, NPHI, PEF	RHOB linear; NPHI reversed; PEF 0-10 barns/electron.	Porosity and lithology.
Track 4	DT, DTS, DTC, shear sonic if available	DT often 40-140 us/ft or 40-240 us/ft. Slowness linear.	Porosity, synthetic seismograms, rock mechanics, seismic tie.

Lithology / Shale-Sand Quick-Look Display

A compact geology review display for lithology screening and correlation.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, SP	GR 0-150/200 API; SP linear mV.	Sand/shale discrimination.
Track 2	Caliper, bit size, borehole flags	Linear inches or mm.	Borehole washout and data quality.
Track 3	Simple lithology fill or Vsh	Vsh 0-1 or 0-100%.	Quick lithology interpretation.

Reservoir Evaluation Display

A calculated/interpreted display combining raw logs and petrophysical outputs.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, caliper, bit size	GR 0-150/200 API.	Quality and lithology context.
Track 2	Deep/medium/shallow resistivity	Log scale.	Saturation and fluid type.
Track 3	Density/neutron/sonic porosity	Porosity 0-45% or 0.45-0.00; neutron often reversed.	Porosity comparison.
Track 4	PHIE, PHIT, Vsh	0-1 or 0-100%.	Reservoir quality.
Track 5	Sw, BVW, hydrocarbon pore volume	Sw 0-1 or 0-100%.	Pay evaluation.
Track 6	Net reservoir / net pay flags	Binary or categorical flag track.	Cutoff-based pay summary.

Density-Neutron Lithology Display

Focused display for porosity, lithology, density correction, and gas-effect review.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR	0-150/200 API.	Lithology context.
Track 2	RHOB + NPHI	RHOB 1.95-2.95 g/cc. NPHI 45 to -15 p.u. reversed.	Porosity, gas crossover, lithology.
Track 3	PEF, DRHO	PEF 0-10. DRHO often -0.25 to +0.25 g/cc.	Lithology and density correction quality.

Resistivity-Focused / Invasion Display

A display for wells with many resistivity curves or multiple depths of investigation.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, SP	GR linear; SP linear.	Lithology reference.
Track 2	MSFL, RXO, SFLU or shallow resistivity	Log scale.	Flushed zone and invasion.
Track 3	ILM, ILD, LLS, LLD, ATxx, RT	Log scale.	True/deeper formation resistivity.
Track 4	Conductivity or invasion ratio	Conductivity linear or ratio scale.	Thin beds, invasion profile, saturation review.

Sonic / Seismic Tie Display

The key WLV-to-seismic bridge display.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, caliper	GR 0-150/200 API; caliper linear.	Lithology and hole quality.
Track 2	DT/DTC, DTS	DT 40-140 or 40-240 us/ft; DTS wider depending tool.	Acoustic properties.
Track 3	RHOB	1.95-2.95 g/cc or local range.	Impedance input.
Track 4	AI, VP/VS, Poisson ratio	Computed ranges, linear.	Seismic tie and rock physics.
Track 5	Synthetic seismogram / reflectivity	Wiggle or variable-area display.	Well-to-seismic correlation.

Formation Tops / Stratigraphic Correlation Display

A correlation-focused display with tops and stratigraphic markers.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR	0-150/200 API.	Correlation marker.
Track 2	Deep resistivity	Log scale.	Reservoir/non-reservoir contrast.
Track 3	Density/neutron or sonic	Conventional porosity or sonic scales.	Lithology/porosity support.
Overlay	Formation tops, zones, sequence boundaries	Horizontal markers and labels.	Stratigraphic interpretation.

Borehole Image Log Display

Image-log display for unwrapped borehole imagery and interpreted features.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, caliper	Linear.	Context and hole quality.
Track 2	Static image	0-360 degree unwrapped borehole image.	Bedding, fractures, texture.
Track 3	Dynamic image	0-360 degree normalized image.	Fine features, fractures, sedimentary fabric.
Track 4	Dip/fracture picks	Tadpoles, sinusoid picks, azimuth/dip.	Structural interpretation.
Track 5	Facies/lithology	Categorical color column.	Geological interpretation.

Nuclear Magnetic Resonance Display

NMR-specific display for pore-system and fluid-mobility review.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, caliper	Linear.	Context and borehole quality.
Track 2	NMR porosity, free fluid, bound fluid	0-45% or 0-0.45.	Porosity partitioning.
Track 3	T2 distribution	Logarithmic T2 time scale.	Pore-size distribution.
Track 4	Permeability estimates	Log scale, mD.	Reservoir quality.
Track 5	Fluid volumes	0-1 or 0-100%.	Movable vs bound fluids.

Geochemical / Elemental Spectroscopy Display

Mineralogy and elemental analysis display.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, spectral GR, K, Th, U	API plus separate channel scales.	Radioactivity source and shale typing.
Track 2	Si, Ca, Fe, S, Al, Mg, K, Ti	Weight percent or normalized yield.	Elemental composition.
Track 3	Quartz, calcite, dolomite, clay, pyrite	0-100% stacked fill.	Mineral model.
Track 4	TOC, brittleness, clay volume	0-100% or local engineering scales.	Unconventional reservoir evaluation.

Dipmeter / Structural Display

Structural display for bedding and fracture orientation.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, caliper	Linear.	Lithology and hole quality.
Track 2	Pad curves / microresistivity	Linear or log depending source.	Bedding response.
Track 3	Dip tadpoles	Dip angle 0-90 degrees; azimuth 0-360 degrees.	Structural dip and bedding.
Track 4	Structural zones/facies	Categorical.	Interpretation.

Mud Log / Drilling Log Composite

Mixed curve, event, and text display for drilling and show review.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	ROP, WOB, RPM, torque	Engineering scales, usually linear.	Drilling behavior.
Track 2	Total gas, C1-C5	Often log or semi-log.	Hydrocarbon shows.
Track 3	Lithology descriptions	Categorical/text.	Geological sample description.
Track 4	GR or MWD curves	Linear.	Correlation with wireline/LWD.
Track 5	Shows, fluorescence, cut, mud losses	Flags/text.	Drilling and reservoir indicators.

LWD / MWD Drilling Display

Real-time drilling and formation-evaluation display.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, azimuthal GR	0-150/200 API.	Geosteering and correlation.
Track 2	Resistivity, phase/attenuation, deep directional resistivity	Log scale.	Bed boundary detection and reservoir/fluid indication.
Track 3	Density/neutron, caliper/standoff	Conventional porosity and caliper scales.	Formation evaluation while drilling.
Track 4	Inclination, azimuth, dogleg severity	Linear engineering scales.	Wellbore trajectory.
Track 5	Toolface, shock, vibration	Engineering scales.	Drilling quality and tool health.

Cased-Hole Cement Bond Display

Cement evaluation display combining correlation, amplitude, and waveform tracks.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, CCL	GR linear; CCL event/curve track.	Depth correlation.
Track 2	CBL amplitude	mV, linear.	Cement-to-casing bond quality.
Track 3	VDL waveform	Waveform image/wiggle.	Cement-to-formation bond and arrivals.
Track 4	Bond index / cement map / impedance	0-1 or categorical/continuous image.	Cement evaluation.

Cased-Hole Saturation / Pulsed Neutron Display

Behind-casing saturation-monitoring display.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, CCL	Linear/event.	Depth correlation.
Track 2	Sigma, capture cross-section	Linear.	Saturation monitoring.
Track 3	C/O ratio, near/far detector ratios	Linear or ratio scale.	Oil/water/gas indication.
Track 4	Computed Sw, gas flags, remaining oil	0-100% or flag tracks.	Reservoir monitoring.

Production Logging Display

Cased-hole flow and production-diagnosis display.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, CCL	Linear/event.	Depth correlation.
Track 2	Spinner / flowmeter	Linear, often centered for up/down flow.	Flow contribution.
Track 3	Temperature	Linear; differential temperature may be added.	Entry points, leaks, crossflow.
Track 4	Pressure	Linear.	Reservoir/wellbore pressure behavior.
Track 5	Fluid density, capacitance, holdup	Linear 0-1 or engineering units.	Phase identification.
Track 6	Perforations, completion schematic	Categorical/mechanical.	Completion context.

Well Integrity / Casing Inspection Display

Mechanical integrity display for casing condition and defects.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, CCL	Linear/event.	Correlation.
Track 2	Internal radius / caliper arms	Inches/mm, linear.	Casing deformation.
Track 3	Metal loss / thickness	Linear or percentage.	Corrosion/wear.
Track 4	Ultrasonic/electromagnetic image	Circumferential image.	Defect localization.
Track 5	Joint/collar/casing schematic	Event/categorical.	Mechanical reference.

Core-Log Integration Display

Display for calibrating logs against core, plugs, facies, and photos.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, density/neutron	Conventional scales.	Log-core correlation.
Track 2	Core gamma, plug porosity, plug permeability	Porosity linear; permeability log scale.	Calibration.
Track 3	Core description / lithofacies	Text/categorical.	Geological truthing.
Track 4	Core photos	Image track.	Visual reference.
Track 5	Sample points	Event symbols.	Plug/sample locations.

Completion / Perforation / Intervention Display

Wellbore schematic plus log overlay for completion and intervention work.

Track position	Curves / data	Recommended scaling	Typical usage
Track 1	GR, CCL	Correlation.	Depth matching.
Track 2	Completion schematic	Tubing, casing, packers, plugs.	Well configuration.
Track 3	Perforations, stages, treatments	Event intervals.	Completion review.
Track 4	Production or pressure/temperature curves	Engineering scales.	Intervention diagnostics.
Track 5	Notes/events	Text intervals.	Operational history.

3. General Positioning Rules

Rule	Recommendation
Depth track	Keep depth fixed and central or left-center. All tracks must align to it.
Lithology/context	Place GR/SP/caliper near the left. This is the geological anchor.
Resistivity	Place resistivity after GR and before porosity. Use logarithmic scaling.
Porosity	Plot density and neutron together. Reverse neutron scale for conventional petroleum displays.
Sonic	Give sonic its own track unless screen width is constrained.
Images	Borehole images, VDL, raster logs, CGM, and TIFF should be image/waveform tracks, not ordinary curve tracks.
Interpretation	Place interpreted products to the right of raw measurements.
Mechanical/completion	Place casing/perforation/completion schematic near cased-hole or production tracks.
Flags	Net pay, facies, zones, and QC flags should use narrow categorical tracks.

4. Common Curve Scaling Recommendations

Curve family	Typical curves / mnemonics	Default scale recommendation
Gamma ray	GR, SGR, CGR	Linear, 0-150 API default; allow 0-200 or custom.
Spectral gamma	K, TH, U, CGR, SGR	Linear; separate units/scales per channel.
SP	SP	Linear mV; allow reversed display.
Caliper	CALI, HCAL, DCAL	Linear inches/mm; overlay bit size.
Bit size	BS, BITSIZE	Linear, same scale as caliper.
Resistivity	RT, ILD, ILM, LLD, LLS, MSFL, RXO, AT10/30/90	Logarithmic. Default 0.2-200 ohm-m; support 0.2-20 and 0.2-2000.
Density	RHOB, RHOZ	Linear, 1.95-2.95 g/cc default; allow custom.
Density correction	DRHO	Linear, often -0.25 to +0.25 g/cc.
Neutron	NPHI, TNPH, CNPOR	Linear porosity; conventional display often 45 to -15 p.u. reversed.
Sonic slowness	DT, DTC, DTS	Linear; common DT 40-140 or 40-240 us/ft.
PEF	PEF, PE	Linear, often 0-10.
Porosity	PHI, PHIE, PHIT	Linear, 0-0.45 or 0-45%.
Water saturation	SW, SWT	Linear, 0-1 or 0-100%.
Shale volume	VSH, VCL	Linear, 0-1 or 0-100%.
Permeability	K, PERM	Log scale if wide range; mD.
Pressure	PRESS, P	Linear engineering scale.
Temperature	TEMP	Linear; support differential temperature.
Spinner / flow	FLOW, RPM, SPIN	Linear, often centered if bidirectional.
CCL	CCL	Event/curve track for collar correlation.
VDL	VDL waveform	Waveform/image display, not simple line curve.
Borehole image	FMI, UBI, XRMI, image raster	0-360 degree unwrapped image track.
Dip	DIP, AZI, tadpoles	Tadpole/symbol track, not ordinary line curve.

5. WLV Implementation Notes

Auto-build logic: The auto-builder should identify curve families first, then select the best representative curves for each family. It should not plot every matching mnemonic.

Resistivity selection: When many resistivity curves exist, classify by investigation depth: flushed/shallow, medium, and deep/true. Use one or a small number of representative curves from each class.

Template selection: Template choice should depend on available curve families and declared workflow context: open-hole, cased-hole, production logging, image log, NMR, geochemical, or seismic tie.

Track object model: Treat curves, images, waveforms, categorical fills, flags, events, annotations, and schematics as separate track object types.

Computed curves: Computed or interpreted curves such as Vsh, PHIE, Sw, net pay, impedance, and facies should be visually distinct from raw measurements.

Scaling: Use sensible defaults but preserve per-project and per-template overrides. Logarithmic vs linear scale must be explicit in the track contract.

CGM/TIFF/DLIS readiness: CGM/TIFF raster logs, VDL waveforms, borehole images, and DLIS-origin data require non-LAS rendering paths. Do not force them into the LAS curve model.

6. Reference Notes

- KGS public well-log interpretation material describes common triple-combo use built around gamma ray, porosity, and resistivity displays.
- SLB public interpretation material describes common presentations combining SP, gamma ray, resistivity, neutron, and density curves.
- EPA cement-bond guidance discusses gamma ray and casing collar locator use for depth correlation in cement bond logging.
- Production logging tool-suite examples commonly include GR/CCL, pressure, temperature, capacitance/density/holdup, and flowmeter measurements.